



Teacher	Mr. Vergel L. Jandayan	Date	January 2019
School	Mataas na Paaralang Neptali A. Gonzales	Time	9:00am to 10:00
Subject	Physical Science	Quarter	2 nd Semester, 3 rd Quarter

GROUP 1 - Naphthalene vs. Water vs. Alcohol

I. Objectives:

Correlate intermolecular forces of attraction and physical properties such as solubility.

II. Materials:

Powdered Naphthalene	Water
100ml of 40% Alcohol	Denatured Alcohol
100ml of 70% Alcohol	Graduated Cylinder
Laboratory spoon	Stirring Rod

III. Procedures

1. Prepare the following materials in separate container each
2. Prepare 50ml of water in the graduated cylinder. Make sure to label the cylinder #1 using masking tape.
3. Put at least 1 teaspoon of powdered Naphthalene in the graduated cylinder of water, shake well and stir using stirring rod, observe.
4. Prepare 70ml of 40% of ethyl alcohol in Cylinder #2, 70ml of 70% ethyl alcohol in Cylinder #3 and another 70ml of denatured alcohol in Cylinder #4. Make sure that each test tube will properly labeled.
5. Put at least 1 teaspoon of powdered naphthalene in the test tube #1, 2 and 3. Shake well using stirring rod for 10 seconds. Observe what will happen.
6. Illustrate your observations.

Guide Questions

1. What happened to the water and naphthalene when mixed together?

2. Why does alcohol can dissolve naphthalene powder while water cannot?

3. How do 40% and 70% of ethyl alcohol differ from denatured alcohol in dissolving naphthalene powdered after mixing up to each other?



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GROUP 2 – Liquid vs. Liquids

I. Objectives:

Correlate intermolecular forces of attraction and physical properties such as solubility.

II. Materials:

500 ml cleared empty bottle	Honey
Dishwashing liquid	Water
Cooking oil	Food color
Beaker	

III. Procedures

1. Prepare cleared 500ml empty bottle and make sure to remove the label in observation of the experiment.
2. Pour at least 100ml of honey/syrup to the beaker #1.
3. Pour at least 100ml of dishwashing liquid to the beaker#2
4. Dilute 100ml of water and food color to the beaker #3.
5. Prepare at least 100ml of cooking oil in the beaker#4
6. Place beaker #1 (honey/syrup) first in the cleared empty bottle.
7. Gently pour beaker#2 (dishwashing liquid) inside the bottle.
8. Add beaker #3(colored water solution) in the bottle.
9. Gently pour beaker #4 (cooking oil) in the bottle as a final step. Then observe.
10. Illustrate your observations.

Guide Questions

1. What happened to the liquid substances?

2. Is there any substance mixed together after the procedure?

3. Why concentration of each liquid substance affects the solubility of the substances?



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GROUP 3 - The Power of SOAP

I. Objectives:

Correlate intermolecular forces of attraction and physical properties such as solubility.

II. Materials:

Cooking oil	Test tube	Test tube rack	Medicine Dropper
Detergent soap	Stirring rod	Water	Cork stopper

III. Procedures

1. Prepare for at least 10ml of water in the test tube #1. Make sure to label the test tube (Oil & Water)
2. Add at least 5 ml of cooking oil using medicine dropper in the test tube of water. Shake well and then observe.
3. Prepare for at least 10ml of water in the test tube #2 and then add 5ml of cooking oil using medicine dropper.
4. Prepare 1 table spoon of detergent soap and then add to test tube #2. Cover the test tube using cork stopper, shake well and observe.
5. Prepare for at least 10ml of water in the test tube #3 and then add 2ml of red water solutions. Shake gradually to form another solutions.
6. Add at least 5ml of cooking oil using medicine dropper to test tube #3. Cover the test tube using cork stopper, shake well and observe.
8. Make sure that all the test tubes are properly label.
9. Complete the table below.

# of Trials	Substance	Combine substance	Reactions
1			
2			
3			

Guide Questions

1. What happened to the water after adding cooking oil in the test tube #1?

2. Why the cooking oil does not mix into the water?

3. How does soap affects the intermolecular force of attraction between water and cooking oil in test tube #2?

4. How will you differentiate the reactions of test tube # 1 and test tube #3?



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GROUP 4 – Acetone and water particles

I. Objectives:

Correlate intermolecular forces of attraction and physical properties such as solubility.

II. Materials:

Acetone 100ml	Test tube rack	Cork stopper	Food color blue and yellow
Beaker	Test tube	Salt	Water
Medicine dropper	White rags		

III. Procedures

Trial 1

1. Prepare at least 200ml of acetone in the beaker, and then dilute it using blue food color.
2. Prepare at least 150ml of yellow water solutions using food color in a beaker.
3. Pour the 150ml of yellow H₂O solutions in a cleared container #1 set as trial 1. Make sure to properly label.
4. Add the 200ml of blue acetone in cleared container #1 and then observe.
5. After few seconds of observations. Cover the cleared container using its cap and then shake well so that the blue acetone will mix to the yellow H₂O solutions. Observe and record your data.
6. Complete the table below

Observation Table for Trial 1

Color of Acetone	Color of H ₂ O Solutions	Color of liquids after shaking

Trial 2

1. Prepare at least 200ml of acetone in the beaker, and then dilute it using red food color.
2. Prepare at least 150ml of blue water solutions using food color in a beaker.
3. Pour the 150ml of blue H₂O solutions in a cleared container #2 set as trial 1. Make sure it is properly label.
4. Add the 200ml of red acetone in cleared container #2 and then observe.
5. After few seconds of observations. Cover the cleared container using its cap and then shake well so that the red acetone will mix to the blue H₂O solutions. Observe and record your data.
6. Complete the table below.

Observation Table for Trial 2

Color of Acetone	Color of H ₂ O Solutions	Color of liquids after shaking

Guide questions:

1. What happened to the acetone and water solutions after adding to the test tube?

2. Which of the two liquids is denser? Why?

3. How do you compare the color of trial 1 and 2 after mixing it together?
